

VOLUME 5

Maintenance and Troubleshooting

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-005
Revision	B
Generated	2026-08-07
Source commit	a3730f9
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Maintenance authority

Use Mazak 413m033 for retained machine maintenance, 41434WB for OEM electrical wiring, YM2V39L for original sequence meaning, and the exact Mitsubishi manual for each installed drive. This retrofit volume covers only added LinuxCNC/Mesa equipment and interfaces. It cannot replace model-specific service instructions.

2. Inspection schedule

Interval	Checks
Before each use	Leaks, loose tools, Z support/brake state, E-stops/guards, LinuxCNC faults, packet/watchdog state, abnormal noise
Weekly	Cabinet filters/seals, fan operation, terminal discoloration, relay indicators, 24 V rails, cable abrasion, coolant intrusion
Monthly	PE bonds, shield drains, connector retention, 7i44/7i84U links, output relays/suppression, log/backup completeness
Quarterly	E-stop/guard functional checks per approved matrix subset, brake holding trend, bus discharge trend, resolver/noise survey comparison
Annually	Full safety validation, torque/terminal inspection per OEM specs, resolver amplitude/phase baseline, analog zero, NIC/latency requalification after updates
After any change	Firmware hash/IDROM, HAL pins, authority validator, control validator, affected hold point and fault-injection tests

3. Safe diagnostic order

- Preserve the first fault: messages, indicator states, HAL values, scope trace, and physical relay state before reset.
- Separate control indication from physical energy. A false HAL pin does not prove a contactor, brake, or bus is safe.
- Isolate and measure the shared P/N bus before opening drive wiring.
- Use the authority CSV to identify the exact card/channel and status; stop if HOLD_CONFLICT or UNBOUND.
- Test from field device toward HAL for inputs, and from reviewed request logic toward load for outputs. Never bypass interlocks in the active machine config.
- After repair, repeat every safety/functional acceptance affected by the changed component or conductor.

4. Motion and resolver troubleshooting

Symptom	Likely areas	Required action
Position noisy at rest	Resolver pair/shield, dual excitation, low return, switching noise	S-ON off; scope differential RESDRV/SIN/COS; compare baseline; operate nearby relays one at a time
Wrong feedback direction	Signed resolver scale or phasing	Return to D8 manual direction proof; do not negate PID gain
Axis moves at zero command	Analog offset, enable/gate error, drive null	E-stop; open S-ON; measure AOOUT and pid/pwmgen values; require <=50 mV before retry
Following error at constant speed	OUTPUT_SCALE/FF1 mismatch, drive gain, resolver scale	Return to volts-per-speed calibration; do not simply raise ERROR
Oscillation/whine	P too high, TRA inner loop, resolver noise	Rollback to last logged settings; inspect drive loop and feedback before retuning
Single drive fault drops group	Shared READY chain behavior	Expected design premise but machine-specific propagation must be logged; inspect all axes and bus
Z drops on stop	Brake sequencing/holding failure	Support Z, isolate power, quarantine motion; investigate hardware timing and brake capacity

5. Ethernet, FPGA, and smart-serial troubleshooting

Symptom	Checks	Do not
7i80HDT unreachable	Dedicated NIC address/link, cable, board power/IP, routing	Do not flash firmware as a connectivity test
packet-error-level rises	NIC driver/IRQ/load, cable, coalescing/offload A/B record, servo latency	Do not increase limits until the cause is understood
watchdog.has_bit	HostMot2 scheduling, thread overrun, Ethernet/FPGA loss	Do not auto-reset into motion
7i84U missing	7i44 channel cable/pinout, remote power/W1, config mask, IDROM	Do not assume device tag or sserial mask
Remote outputs drop	Remote watchdog time, link, field bank voltage, VIN, output fault	Do not disable the watchdog
HAL pins differ	Loaded bitfile/readhmid/config string	Do not rename around an unproven firmware mismatch

6. Spindle and orient troubleshooting

Symptom	Checks	Safe response
No spindle rotation	Static/dynamic permits, READY, fault, FWD/REV/RUN, AOUT3, gear confirm	Keep low energy; verify exact FR-SX terminal states
Wrong direction	FWD/REV exclusivity and analog sign/mode	Drop all spindle permits; correct one documented sign stage
No orient arrival	Zero speed, gear, ORCM1, detector/parameter mode, ORA1 polarity	Allow component timeout; no automatic retry
Orient cancel restarts spindle	Run bit remained asserted; adjacent-family behavior	Treat as hazardous until exact FR-SX command order is proven
At-speed unreliable	IN13 source, polarity, threshold, debounce	Do not bypass ATC spindle-state interlock

7. ATC and field-I/O troubleshooting

Symptom	Checks	Safe response
Magazine position wrong	IN19-IN23 BCD, IN28 validity, pocket/tool table	Stop indexing; compare each bit at a known pocket
Clamp state disagrees	PRS-8/9, pressure, SOL-10 topology, OUT9/10	Support tool; isolate hydraulics; resolve valve mapping
Gear does not confirm	Zero speed, OUT7/OUT8, PRS-10/12, coil identity	Do not spin; HOLD_CONFLICT remains until traced
Output does nothing	Field bank/VIN, W1, watchdog, interposing relay, coil voltage/current	Measure at each boundary; never apply 100 VAC to 7i84U
Unexpected adjacent input	Commoning, sensor wiring, input mode, induced noise	Do not promote to HAL_VERIFIED; trace and repeat
ATC stalls	Current state, prerequisites, timeout/alarm, hydraulic pressure	No recovery jog; isolate and use approved manual recovery

8. Backup, restore, and rollback

- Maintain an immutable original-control photo/wire ledger and M-2 parameter/ladder backups before removing conductors.
- For every released retrofit state, archive the git commit, authority CSV, INI/HAL, custom components, tool table, bitfile binary/hash/source, readhmid, HAL pin dump, host package/kernel/NIC record, and signed test evidence.
- Tag a known-safe baseline only after its hold-point evidence is complete. A git tag without physical acceptance is not a safe machine baseline.
- After a fault during commissioning, return both output holds FALSE, physically isolate S-ON, restore the logged low-authority configuration, and preserve all evidence before retry.
- Reverting to the original M-2 requires the removed-conductor ledger and independent continuity check; never reconnect from memory or an old Rev A PDF.

9. Spares and obsolescence

Mitsubishi TRA/HD and FR-SX hardware is obsolete. Record installed suffixes before purchasing used spares. Store repaired/used units with test reports and retain parameter/trim positions. Mesa spares must match board revision and verified firmware package; a replacement 7i80HDT must not be flashed from an assumed filename.

10. Required maintenance records

Record	Location
I/O checkout and state measurements	wiring/io_checkout_sheet.md + docs/commissioning_logs
Resolver scope/phasing/scale	docs/commissioning_logs/resolver_traces
Servo tuning and faults	docs/commissioning_logs/tuning and per-axis logs
Bus discharge/regen	docs/commissioning_logs/dc_bus_discharge_*
E-stop fault matrix	docs/commissioning_logs/estop_validation_*
Network/latency/cable-yank	docs/commissioning_logs/latency_*
Firmware package	mesa/firmware with SHA256SUMS/readhmid/HAL pins
Configuration release	Git commit/tag plus manual-set manifest.json